PROJECT PRESENTATION II

Academic Text Style Enhancement

A Transformer + Mamba hybrid for meaning-preserving revision

Milestone: real data, working code and initial GPU checks

Borui Ning Licheng Wang Haoxiang Ning Lingfeng Lu

Week 9 October 25–31, 2026

Dataset

IteraTeR-HUMAN sentence release

4,018 original records 676 retained arXiv revision pairs

Split	Pairs	Documents
Train	491	76
Validation	98	9
Test	87	10

Real human revision

Source: Discovering new intents is a crucial task in a dialogue system .

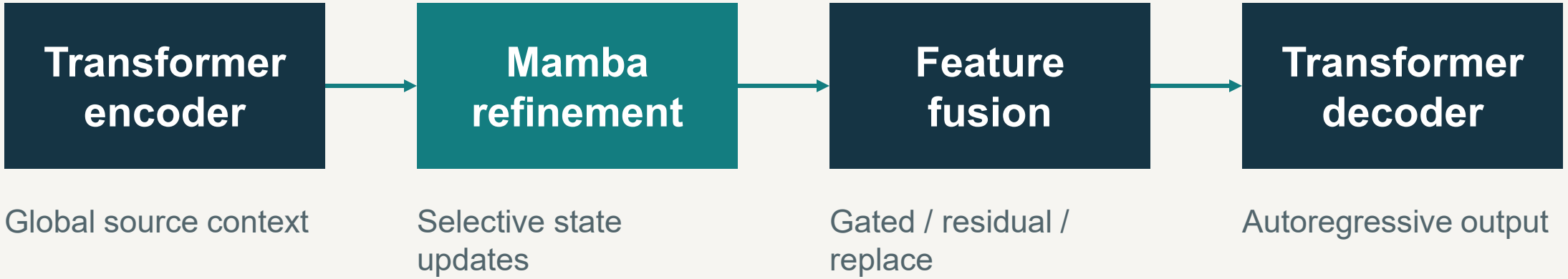
Reference: Discovering new intents is a crucial task in dialogue systems .

Document-group split, seed 42. Cross-split document and exact-text overlap: 0.

Proposed model

Sentence-level detection and rewriting with author review

Source tokens



$$Z = H + G \odot M \qquad G = \text{sigmoid}(W[H; M] + b)$$

Objective: target-token cross-entropy with padding ignored

Current adapter: sequential PyTorch reference. Official optimized kernel validation is pending.

Current progress

Both pipelines completed forward, backward and generation

30 GPU updates per model 64 train pairs 16 validation pairs

Model	Initial batch CE	Final batch CE	Validation CE
Transformer	7.778	6.571	6.841
Hybrid	7.778	6.570	6.840

4 integrity tests passed. Checkpoints and logs saved locally.

Random tiny models verify connectivity. Rewriting quality remains unestablished.

RTX 4060 Laptop GPU. PyTorch 2.2.0+cu121. Test set excluded from model evaluation.

Current challenges

Data coverage and model validation remain the main constraints

Problem	Current evidence	Response
Style supervision	24 of 676 pairs have a style label	Expand academic pairs and annotate tone independently
Meaning preservation	Number checks miss negation, hedging and entities	Audit source/reference pairs and use blind human review
Mamba integration	Reference scan runs. Official backend is untested	Validate Linux/WSL2 kernels, gradients and memory
Prototype maturity	Random tiny models and a heuristic detector	Fine-tune pretrained BART and train a labeled detector

No final quality, speedup or academic-style improvement claim is supported yet.

Next steps

Suggested team targets after the milestone

Target date	Owner	Reviewable output
Oct 25–31	Licheng Wang	Audit 50 pairs and create detector labels
Nov 1–7	Haoxiang Ning	Pretrained BART baseline and validation checkpoint
Nov 8–14	Borui Ning	Official Mamba checks and fusion ablations
Nov 15–21	Lingfeng Lu + team	Blind review, efficiency table and author-review demo

Matched splits and decoding. Validation-based selection. Test after protocol freeze.

Working targets, not course-issued deadlines. Confirm the final course schedule.

Milestone summary

Implementation evidence

676 real academic revision pairs with auditable splits

Working Transformer and Hybrid training pipelines

Saved GPU logs, checkpoints and reproducible code

Next focus: pretrained training and independent quality evaluation

Local repository prepared. Remote GitHub publication remains pending.